ثانوية المدينة التعليمية
Education City High School

1.3

Rates of Change in Linear and Quadratic Functions

Student Notes and Guided Practice

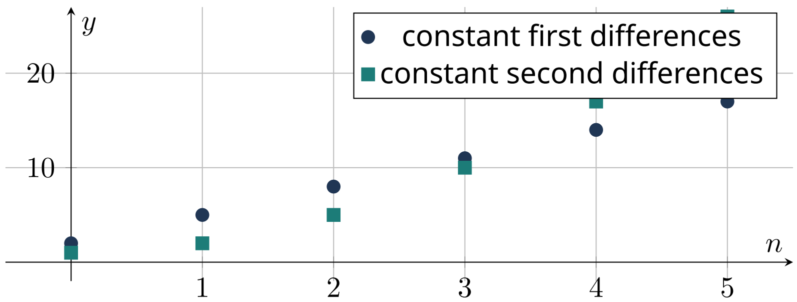
AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Distinguish constant first differences from constant second differences.
- Connect linear rate behavior to constant slope and quadratic rate behavior to a linear rate pattern.
- Use rates of change to identify and compare linear and quadratic functions.

Essential Question

How do first and second differences distinguish linear from quadratic change?



Concept Framework

Key Idea

Equal input steps are essential when using finite differences.

Key Idea

Linear functions have constant first differences and zero second differences.

Key Idea

Quadratic functions have constant nonzero second differences.

Key Idea

For $f(x) = ax^2 + bx + c$ with step size h, the constant second difference is $2ah^2$.

Formula Map

Formula and Structure

$$\Delta f = f(x+h) - f(x)$$

Formula and Structure

$$\Delta^2 f = \Delta f(x+h) - \Delta f(x)$$

Formula and Structure

$$\Delta^2(ax^2 + bx + c) = 2ah^2$$

Worked Examples

Worked Example 1

Problem. The values $4, 9, 14, 19, 24$ occur at consecutive integers. Classify the pattern.
Solution. First differences are all 5, so the values come from a linear function on these inputs.

Worked Example 2

Problem. The values $2, 7, 16, 29, 46$ occur at consecutive integers. Determine the quadratic leading coefficient.
Solution. First differences are $5, 9, 13, 17$; second differences are 4. With $h = 1$, $2a = 4$, so $a = 2$.

Worked Example 3

Problem. Explain why a quadratic can have the same average rate on two different intervals.
Solution. Secant slopes for a quadratic depend on interval midpoint. Intervals with the same midpoint can have the same average rate.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Classify the data at consecutive integer inputs: $2, 7, 12, 17, 22$. Show differences.

2. Classify the data at consecutive integer inputs: $1, 4, 9, 16, 25$. Show differences.

3. Classify the data at consecutive integer inputs: $5, 1, -3, -7, -11$. Show differences.

4. Classify the data at consecutive integer inputs: $3, 8, 17, 30, 47$. Show differences.

5. Classify the data at consecutive integer inputs: $0, 2, 8, 18, 32$. Show differences.

6. A quadratic table at step size 2 has constant second difference 24. Find its leading coefficient.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ____________________.

2. A restriction I must remember is ____________________.

3. One graphical feature I can justify algebraically is ____________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.